

MATHEMATICS FOR ML: RANDOM VARIABLES & DISTRIBUTIONS

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Outline

① RANDOM VARIABLES & DISTRIBUTIONS

About Me

Yogesh Haribhau Kulkarni

Bio:

- ▶ 20+ years in CAD/Engineering software development
- ▶ Got Bachelors, Masters and Doctoral degrees in Mechanical Engineering (specialization: Geometric Modeling Algorithms).
- ▶ Currently doing Coaching in fields such as Data Science, Artificial Intelligence Machine-Deep Learning (ML/DL) and Natural Language Processing (NLP).
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Saturdays, 2 to 5pm
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About Content

- ▶ Bare essential mathematics needed for Machine/Deep Learning
- ▶ Just to get you started.
- ▶ Each field within Machine/Deep Learning can go extremely deep.
- ▶ This is to give you enough foundation needed.

About Me

- ▶ I am not a mathematician.
- ▶ And this course won't turn YOU into one!!! (if you are not already)
- ▶ But, I will surely attempt to make you get interested in the learning, more.

Random Variables

Random Variables

- ▶ Random Variables, is actually a misnomer. They are not variables in the programming sense. They are Functions mapping outcomes of Random Processes to real numbers.
- ▶ Example:
 - ▶ Random Process: Coin flipping
 - ▶ Outcomes: HEAD or TAIL
 - ▶ Real numbers: 1 for HEAD and 0 for TAIL
- ▶ Another Example:
 - ▶ Random Process: rolling of 7 dice
 - ▶ Outcomes: take top numbers of each
 - ▶ Real numbers: Sum of the top numbers
- ▶ Whats the use of doing this?
- ▶ Gives easy way to express the random processes, in an equation.

Random Variables

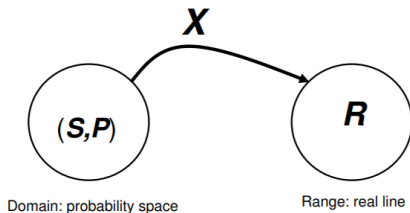
- ▶ Example: Find probability of sum of top of 7 dice being less than 30. If Y is said to represent “sum of top of 7 dice” then it can be expressed as $P(Y \leq 30)$.
- ▶ Similarly, Random variables allows us to ask questions in mathematical way. If we flip 5 coins and want to answers questions like:
 - ▶ What is the probability of getting exactly 3 heads?
 - ▶ What is the probability of getting less than 4 heads?
 - ▶ What is the probability of getting more than 1 head?
- ▶ Then our general way of writing would be:
 - ▶ $P(\text{Probability of getting exactly 3 heads when we flip a coin 5 times})$
 - ▶ $P(\text{Probability of getting less than 4 heads when we flip a coin 5 times})$
 - ▶ $P(\text{Probability of getting more than 1 head when we flip a coin 5 times})$
- ▶ But if we use random variables to represent above questions then we would write:
 - ▶ $P(X = 3)$
 - ▶ $P(X < 4)$
 - ▶ $P(X > 1)$
- ▶ As we can see above random variables makes our task much easier to quantify results of any random process and apply math and perform further computation.

Random Variables

- ▶ Intuitively, those variables which can take different, unpredictable values are called random variables.
- ▶ A Random Variable is different from the variable in algebra as it has whole set of values and it can take any of those randomly. Variable used in algebra cannot have more than a single value at a time.
- ▶ If random variable $X = 0, 1, 2, 3$. Then X could be 0, 1, 2 or 3 randomly where each of them might have a different probability.
- ▶ We use capital letter for random variables to avoid confusion with traditional variables.
- ▶ If the variables are discrete in nature, they are called discrete random variables. Example: In an experiment of tossing 2 coins, we need to find out the possible number of heads. Therefore $X = 0, 1, 2$
- ▶ Similarly, if the variables are continuous in nature, then it is called continuous random variable. Example: temperature in a city lies between 30C and 45C centigrade. The temperature can take any value in the interval 30C to 45C.

Mathematically, Random Variable

- ▶ A (real-valued) random variable, often denoted by X (or some other capital letter), is a function mapping a probability space (S, P) into the real line $\mathbb{R}$
- ▶ Associated with each point s in the domain S the function X assigns one and only one value $X(s)$ in the range $\mathbb{R}$

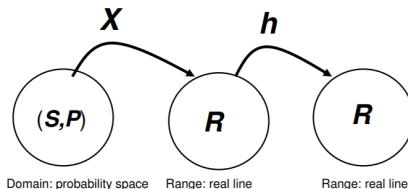


- ▶ As such, a random variable has a probability distribution. P is the probability measure on the underlying sample space S

(Ref: IEOR 4106: Introduction to Operations Research: Stochastic Models)

Function of a Random Variable

- ▶ Suppose that we are given a random variable X mapping the probability space (S, P) into the real line $\mathbb{R}$ and we are given a function h mapping $\mathbb{R}$ into $\mathbb{R}$.
- ▶ Then $h(X)$ is a function mapping the probability space (S, P) into $\mathbb{R}$.
- ▶ As a consequence, $h(X)$ is itself a new random variable, i.e., a new function mapping (S, P) into $\mathbb{R}$



(Ref: IEOR 4106: Introduction to Operations Research: Stochastic Models)

Random variables

A **random variable** is generated by a “random” function.

Types:

- ▶ Nominal
- ▶ Ordinal
- ▶ Quantitative

Nominal Random Variable

- ▶ A **nominal** random variable, also called **qualitative** or **categorical**, takes on values in a set of names or labels.
- ▶ Examples of nominal sample spaces include geographical locations, biological species, and ethnic categories.
- ▶ Data from a nominal random variable cannot be **strictly ordered**.

Ordinal Random Variable

- ▶ An **ordinal** measurement is usually either a **ranking** (1st, 2nd, etc.) or a **rating** (good, bad, favorable, strongly agree, etc.).
- ▶ Ordinal values can be ordered, but do not have **measurement units**
- ▶ It doesn't always make sense to subtract or divide their numerical values.

Quantitative Random Variable

- ▶ A **quantitative** random variable represents a magnitude, for example the result of counting something, or the result of measuring a physical quantity such as length, width, or volume.
- ▶ Quantitative measurements can always be ordered.

Quantitative Random Variable

Types:

- ▶ Discrete: Number of days that it rains yearly
- ▶ Continuous: Amount of rain on a given day

Note that these designations aren't used in the case of nominal or ordinal random variables.

Quantitative Random Variable

- ▶ A **discrete** random variable takes on finitely many values, or infinitely many separated values (e.g. $0, 1, 2, \dots$).
- ▶ A **continuous** random variable can take on any real number in some interval (e.g. $[0, 1]$ or $[0, \infty)$).
- ▶ Gray area between discrete and continuous random variables: e.g. monetary values recorded in dollars or dollars and cents

Units, populations and samples

- ▶ A **unit** (more specifically “**sampling unit**”) is one member of the collection.
- ▶ A **population** is the set of all units of interest.
- ▶ In practice, we cannot observe the whole population, so we usually work with a **sample** of units

Units, populations and samples

- ▶ Properties of a sample will generally differ from properties of the population.
- ▶ For example, the 100 children in our sample may have somewhat higher or somewhat lower average calorie intake than the population at large.
- ▶ Statistics: how samples relate to populations, and what we can confidently state about a population given the limited information in a sample.

Random Variables

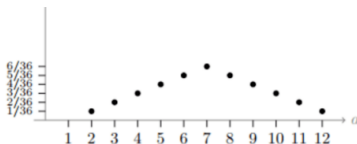
- ▶ Random variable can take on randomly different value.
- ▶ But are all the values that it takes can be equal or likely be equal too, or is it more likely that the random variable take a particular value more often than other?
- ▶ Depends on the way (function) using which random numbers are generated.

Random Variable Generation

- ▶ Experiment: records numbers from a SINGLE six face die.
- ▶ X axis shows 6 possible outcomes.
- ▶ Y axis shows their probabilities.
- ▶ For each of the 6 possible values on X axis, the Y value is same ie $1/6$
- ▶ Uniform distribution

Random Variable Generation

- ▶ Experiment: records sum of numbers from TWO six face dies.
- ▶ X axis shows 12 possible outcomes.
- ▶ Y axis shows their probabilities.
- ▶ For each of the 12 possible values on X axis, the Y value is NOT SAME.



Probability Mass Function

Random Variable Generation

- ▶ If the random variable is discrete in nature, we use Probability Mass Function to describe its probability distribution
- ▶ If the random variable is continuous in nature, we use Probability Density Function to describe its probability distribution.

Introduction to Probability distributions

Probability distributions

The probability distribution for a random variable X gives the possible values for X , and the probabilities associated with each possible value (i.e., the likelihood that the values will occur). Types:

- ▶ Discrete probability distributions
- ▶ Continuous probability distributions

Discrete Probabilities

- ▶ If a random variable has a discrete sample space, a probability distribution is just a listing of the probabilities with which each point in the sample space occurs.
- ▶ These probabilities must sum to 1.

Example: Discrete Probabilities

- ▶ Survey of a political candidate.

x	$P(X = x)$
Favorable	0.3
Neutral	0.5
Unfavorable	0.2

- ▶ The mapping from points in a discrete sample space to the corresponding probabilities is called a **probability mass function (PMF)**.
- ▶ The sample space is finite.

Many Discrete Probabilities

- ▶ Infinitely many rows

x	$P(X = x)$
0	0.37
1	0.37
2	0.18
3	0.06
...	...

- ▶ Although there are infinitely many nonzero probabilities, they still must sum to one.

Probability Mass Function

- ▶ function that generates discrete variables with probabilities
- ▶ $f(x_i) = P(X = x_i)$ is the probability that X has the value x_i .
- ▶ Properties:
 - ▶ $0 \leq f(x_i) \leq 1$
 - ▶ $\sum f(x_i) = f(x_1) + f(x_2) + \dots = 1$

Example: Probability Mass Function

- ▶ Suppose the random variable X is the number of rooms in a randomly chosen owner-occupied housing unit.
- ▶ The distribution of X is:

Rooms (X)	1	2	3	4	5	6	7
Probability	.083	.071	.076	.139	.210	.224	.197

Probability Density Function

- ▶ function that generates continuous random variables with probabilities
- ▶ $f(x_{(a,b)}) = P(X = x_{(a,b)})$ is the probability that X has the values in range (a, b) .
- ▶ Properties:
 - ▶ $0 \leq f(x_i) \leq 1$
 - ▶ $\int f(x_i) = 1$

Distributions of continuous random variables

Two important identities are

$$\begin{aligned}P(X > x) &= 1 - P(X \leq x) \\P(a < X \leq b) &= P(X \leq b) - P(X < a).\end{aligned}$$

Therefore we only need to know $P(X \leq x)$ for all values of x and we can determine the probability that X lies in any given interval.

Its area under curve

Distributions of continuous random variables

Example: Suppose we want to know the probability that a person in a certain population is between 170cm and 180cm tall. If we know that the probability of being less than 180cm tall is 0.8, and the probability of being less than 170cm tall is 0.6, then the probability of being between 170cm and 180cm tall is $0.8 - 0.6 = 0.2$.

To put this in mathematical notation, let X denote the height of a randomly selected person from the population. Then

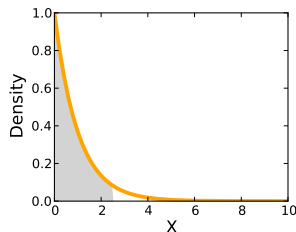
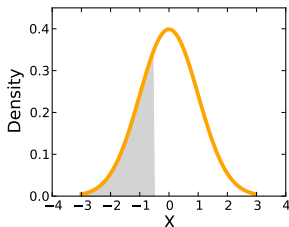
$$P(170 < X \leq 180) = P(X \leq 180) - P(X < 170) = 0.8 - 0.6.$$

Density functions

The distribution of a continuous random variable is described by a **density function**.

A density function is a non-negative function such that the area between the horizontal axis and the function is exactly 1.

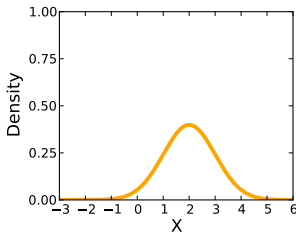
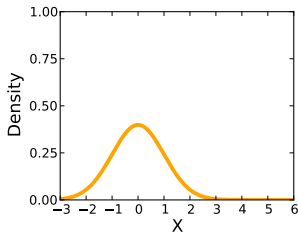
The orange curves below are density functions. The grey area on the left is $P(X \leq -0.5)$, which is 0.31 for this distribution. The grey area on the right is $P(X \leq 2.5)$, which is 0.92 for this distribution.



Characteristics of density functions

A basic property of a density function is its **location** – the central or most typical value.

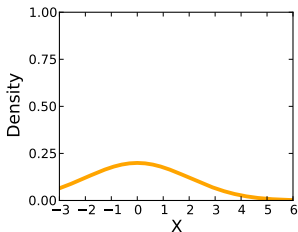
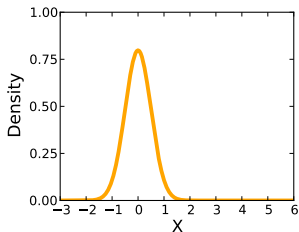
There are many measures of location, but by any reasonable measure the density on the right, below, would have a greater location than the density on the left.



Characteristics of density functions

Another basic property of a density function is its **scale**, referring to spread or width of the distribution.

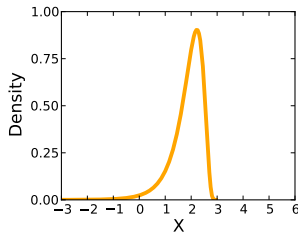
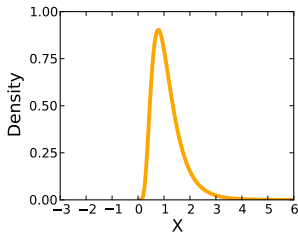
There are many measures of scale, but by any reasonable measure the density on the right, below, would have a greater scale than the density on the left.



Characteristics of density functions

Another basic property of a density function is its **skew**, referring to whether one of the two tails of the density is thicker than the other.

The density on the left, below, is right-skewed, and the density on the right, below, is left-skewed.



Examples of Probability distributions

Discrete Probability Distributions

Some examples:

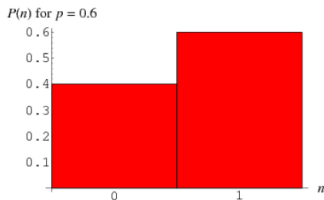
- ▶ Bernoulli distribution : $P(X = 1) = p, P(X = 0) = 1 - p$
- ▶ Binomial distribution : -Two possible outcomes, probability constant = $\binom{n}{x} p^x (1 - p)^{n-x}$
- ▶ Poisson distribution
- ▶ Negative Binomial

Bernoulli

Bernoulli random variable is used when the experiment results in either success or failure where success is represented as 1 and failure as 0. If P is probability of success, then probability mass function of Bernoulli variable x is given as:

$$P(x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \end{cases}$$

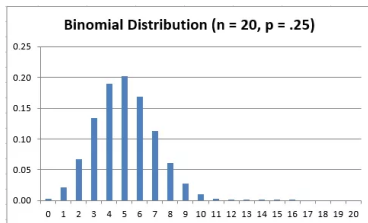
Bernoulli



Binomial

Binomial distribution is used when we want to count how many success we have when we repeat an experiment for n numbers of time independently. The probability mass function of binomial variable x , where p is probability of success, is given as:

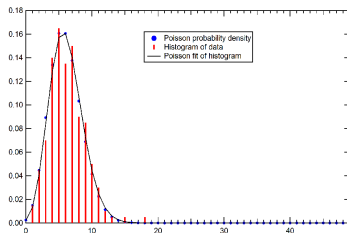
$$P(x) = \binom{n}{x} p^x q^{n-x} = \frac{n!}{(n-x)!x!} p^x q^{n-x}$$



Poisson

Poisson variable x is used when we are counting the number of occurrences of an event in a unit of time such that the occurrences are independent and rarely simultaneous. For an average number of occurrence λ the probability mass function is given as:

$$P(X) = \frac{\lambda^x e^{-\lambda}}{X!}$$



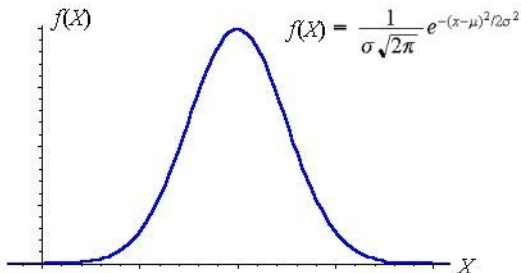
Continuous Probability Distributions

Some examples:

- ▶ Normal distribution
- ▶ Uniform distribution
- ▶ Exponential distribution
- ▶ Weibull distribution

Normal

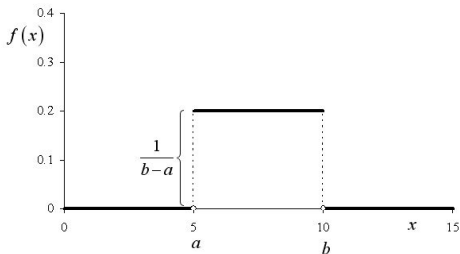
Normal random variable x comes handy for different situations. We can use it to model physical measurements like weight, height etc. We can also use it to model error made by measuring instruments. In general, it can be used when the average and variance of the quantity being measured is known. The probability density function is given as:



Uniform

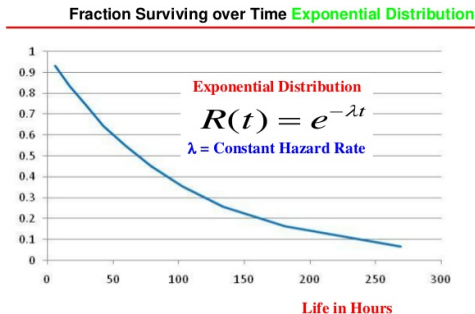
We use uniform random variable x when the probability density for every value between an interval (starting from a and ending at b) is equal. The probability density function of x is given as:

$$f(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$



Exponential

Exponential variable x are used when we are measuring the time until the first occurrence of an event, such that the occurrences in disjoint time intervals are independent and rarely simultaneous. For the average number of occurrences per unit time a , the probability density function is given as:



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Thanks ...

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